

Session 2

Using JSP Directive and Scriptlet



Review

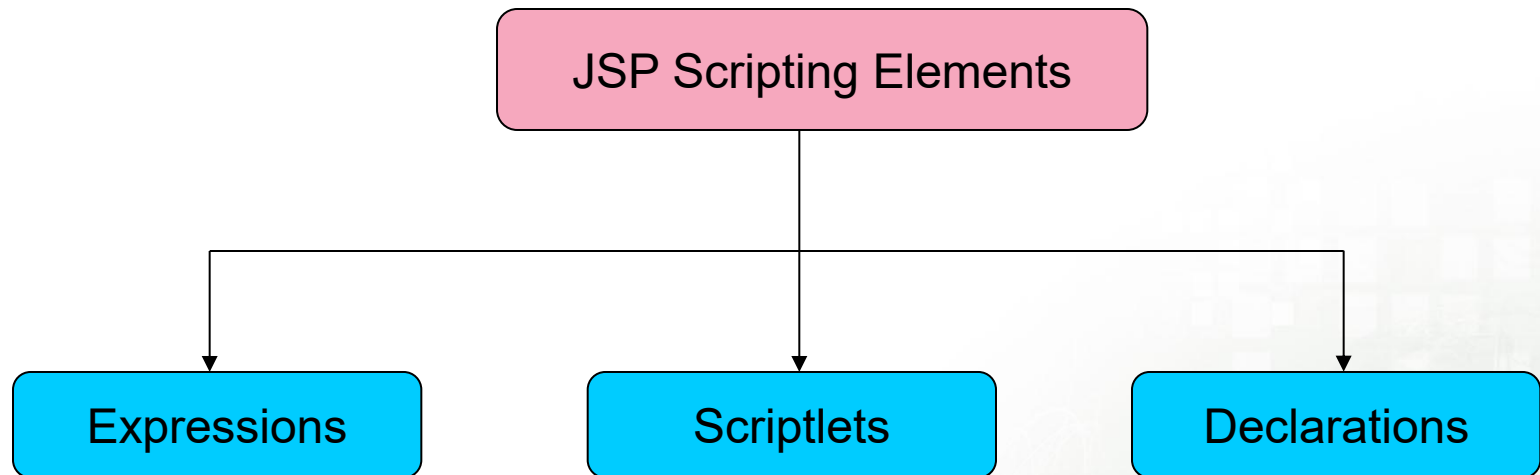
- ❑ JSP stands for Java Server Pages and they are saved with the extension `.jsp`.
- ❑ JSP uses Java Programming language and the set of class libraries available in Java.
- ❑ JSP page uses JavaBeans to present static content on Web page and Java Code to present dynamic content.
- ❑ A JSP page can be created using standard HTML tools such as Macromedia Dreamweaver.
- ❑ JSP uses reusable and cross-platform components like JavaBeans.
- ❑ JSP makes it easy to develop a page by allowing the creation of new user-defined tags.
- ❑ Elements of JSP page are **Static Content, JSP Directive, JSP Expression** and **JSP Scriptlet**.

Objectives

- ❑ *Define a JSP scriptlet*
- ❑ *Explain the use of scriptlets in JavaServer Pages*
- ❑ *Describe JSP directives*

JSP Scripting Elements

- ❑ The JSP scripting elements are:



JSP Expressions

- ❑ JSP expressions are enclosed in tags `<%=` and `%>`.

```
<%@page contentType="text/html"%>
<%@page pageEncoding="UTF-8"%>
<%@page language="java" %>
<html>
  <head>
    <meta http-equiv="Content-Type"
content="text/html; charset=UTF-8">
    <title> JSP1 </title>
  </head>
  <body>
    <h1>Mathematical functions</h1>
    <b>Value of PI : </b><%=Math.PI %><br />
    <b>Exponential Value of 10 : </b><%=Math.exp(10) %><br />
    <b>floor 3.14 : </b><%=Math.floor(3.14) %><br />
    <b>ceil 3.14 : </b><%=Math.ceil(3.14) %><br />
    <b>log 10 : </b><%=Math.log(10) %><br />
    <b>max 10,11 : </b><%=Math.max(10,11) %><br />
    <b>min 10,11 : </b><%=Math.min(10,11) %><br />
  </body>
</html>
```

Expressions to be evaluated

JSP Scriptlet 4-1

- ❑ JSP scriptlets are enclosed in tags `<%` and `%>`.

```
<%@page contentType="text/html"%>
<%@page pageEncoding="UTF-8"%>
<html>
  <head> A Simple JSP Scriptlet </head>
  <body>
    <% int num=0;
    for (num=0;num<5;num++) {
      num=num+1;
      out.println("<br>The value of num is: " +num);
    }
    %>
  </body>
</html>
```

Demonstration: Example 2

JSP Scriptlet 4-2

❑ A scriptlet that generates random numbers

```
<%@page contentType="text/html"%>
<%@page pageEncoding="UTF-8"%>
<html>
  <head>
    <title> JSP1 </title>
  </head>
  <body>
    <h1>Displaying random number pyramid</h1>
    <%
      for(int i=1;i<10;i++) {
        for(int j=0;j<=i;j++) {
          out.println(j);
        }
        out.println(i+"<br/>");
      }
    %>
  </body>
</html>
```

Demonstration: Example 3

JSP Scriptlet 4-3

❑ A scriptlet that displays even numbers between 1 to 10

```
<%@page contentType="text/html"%>
<%@page pageEncoding="UTF-8"%>
<%@page language="java"%>
<html>
  <head>
    <meta http-equiv="Content-Type"
content="text/html; charset=UTF-8">
    <title> Simple Scriptlet </title></head>
    <body>
      <%
        int c=0;
        for(int i=1;i<=10;i++) {
          c=i%2;
          if(c==0) {
            out.println(+i+"  is an even number<br/>");
          }
        }
      %>
    </body>
</html>
```


JSP Scriptlet 4-4

- ❑ A scriptlet that displays the number of tokens and characters in a given sentence.

```
int tokens=0;
Vector chars=new Vector();
StringTokenizer st=new StringTokenizer(str);
tokens=st.countTokens();
out.println("<b>Given string is: </b>"+str+"<br/>");
out.println("<b>Number of tokens in the given
string are: </b>"+tokens+"<br/>");
out.println("<b>Number of characters in the given
string are: </b>"+str.length());

%>
</body>
</html>
```

```
String str="I am a good Programmer";
```

Demonstration: Example 5

JSP Declarations

- ❑ JSP Declarations are enclosed in syntax:

`< % ! Java code %>`

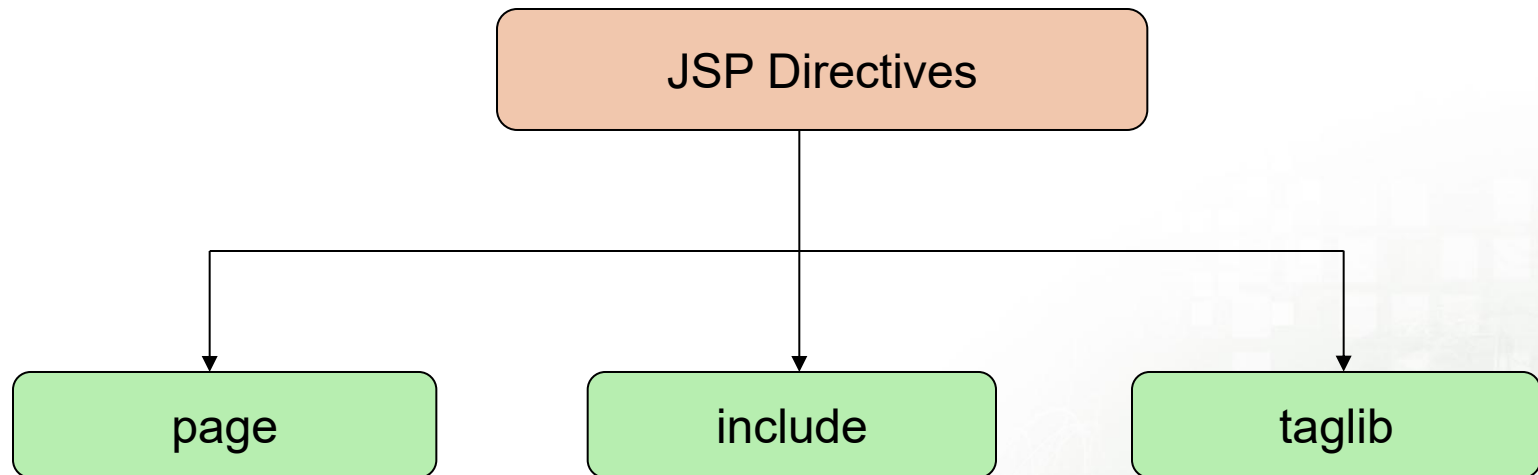
```
<%! int price = 10;
    public int display(int num)
    {
        return(num) ;
    }
%>
```

Variables and methods to be used in Java code are declared using JSP declaration

```
<html>
<body>
<h1>The value of price is <%= price%></h1>
<h1>The value of num is <%= display(5) %></h1>
</body>
</html>
```

JSP Directives

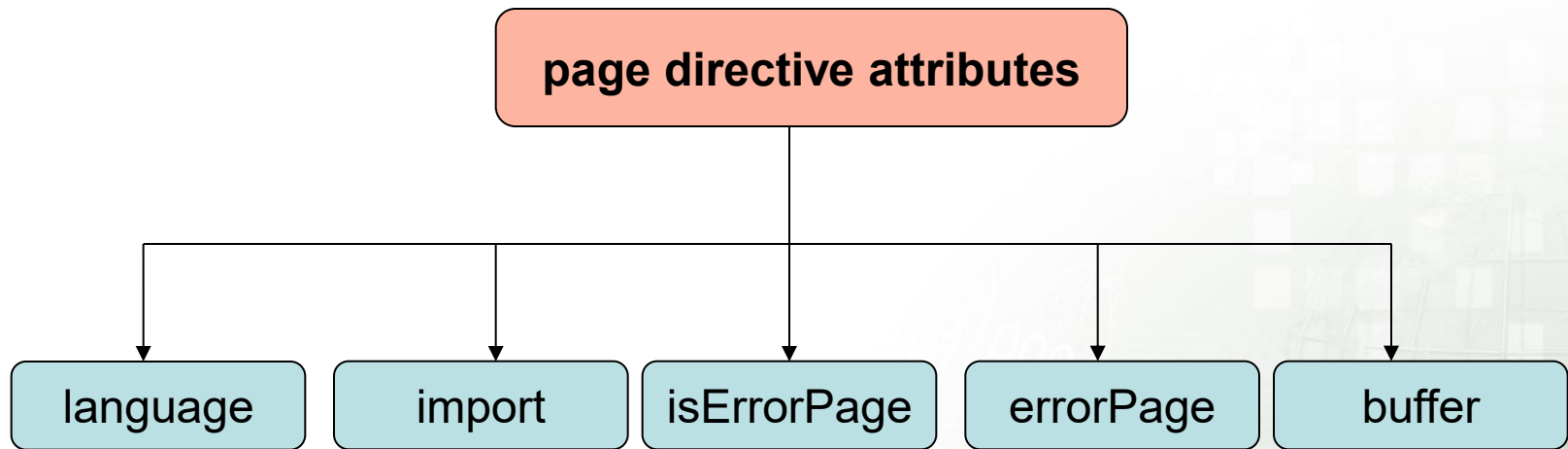
- ❑ The JSP directives are:



Syntax of JSP Directive is `<@ directiveName attribute = "value" %>`

page Directive 2-1

- ❑ page directive is used to set the attributes of a JSP page.



`buffer`– Sets the size of the buffer to store client request

page Directive 2-2

```
<%@ page language = "Java" import="java.util.*"
    buffer="8kb" errorPage="error.jsp" isErrorPage="false" %>
<html>
<head>
<title> Testing Page Directive</title>
</head>
<body>
<h1> This is a test page</h1>
</body>
</html>
```

page directive attributes

include Directive

- ❑ `include` directive is used to include HTML files or another JSP page in a JSP page.

```
<html>
<head> Include Directive Test page </head>
<body>
<%@ include file = "/test.html" %>
</body>
</html>
```

File to be included in a JSP page

Using page and include Directive 3-1

```
//Message.html
```

```
<html>  
<head>  
<H1>Welcome to Shop Stop</H1>  
</head>  
</html>
```

```
//IncludeFile.jsp
```

Scripting language
is set as Java

```
<%@page language="java" %>  
<%@include file="Message.html"%>
```

The contents of Message.html file
are included in IncludeFile.jsp

Demonstration: Example 6 and 7

Using page and include Directive 3-3

The attributes of count variable and requested URI are retrieved using `getAttribute()` method

```
<b>
Include Example
</b>
<%
String counts=(String)session.getAttribute("count");
String requestedURI=(String)session.getAttribute ("requestedURI");
String color="#FFFFCC";
%>
<%
out.println("<h3>Displaying values from the session
object.</h3>");
%>
<b>This page was accessed </b> <%=counts%> <b>times.</b><br
/>
<b>Requested URI: </b> <%=requestedURI%><br />
```

Demonstration: Example 8

Using page and include Directive 3-2

```
<%!  
int count=0;  
%>  
<%  
    session.setAttribute("count",String.valueOf(count));  
    session.setAttribute("requestedURI",request. getRequestURI());  
    if(count==10) {  
        out.println("<b>Congratulations you have just  
            won a surprise gift.</b>");  
    }  
    count++;  
%>  
<%@include file="IncludePage.jsp"%>
```

This example uses `setAttribute()` method to set the attributes of integer variable and requested URI and includes the file `IncludeJsp.jsp`

Demonstration: Example 9

taglib Directive

- ❑ Specifies that the JSP page uses custom tags
- ❑ Specifies the tag library that defines the custom tags
- ❑ Assigns a tag prefix for the custom tags

```
<%@taglib uri="/WEB-INF/tlds/mytaglib.tld" prefix = "mytags" %>
```

uri attribute specifies the location of the tag library descriptor file

Body tag

```
<mytags:tags1 attr1="a" attr2="b">  
body  
</mytags:tags1>
```

Summary 2-1

- ❑ The JSP scripting elements are `Declarations`, `Scriptlets`, and `Expressions`.
- ❑ JSP Expressions are used to evaluate expressions and are written in the form `<% = code %>`.
- ❑ JSP Declarations are used to declare variables and methods and are written in the form `<%! code %>`.
- ❑ JSP Scriptlets are used to insert complex Java code into a JSP page and are written in the form `<% code %>`.

Summary 2-2

- ❑ Directives are of three types: `page` directive, `include` directive, and `taglib` directive.
- ❑ All directives are enclosed in the `<%@ directive %>` tag.
- ❑ The `page` directive is used to set the attributes of a JSP page. The `include` directive is used to include a file into a JSP page. The `taglib` directive allows the user to use custom tags.